



Equations

Atomic equation notes (`\latex-env: equation`) embedded as numbered equations with working cross-references.

This document tests how equations from atomic notes are embedded as numbered equations. Convention: the note's frontmatter carries `\latex-env: equation` and its body contains exactly one `$$...$$` display-math block. The filter wraps that block in `\begin{equation}\label{eq:X}...\end{equation}`. The `caption` frontmatter is optional and for documentation only — it does not render in the PDF.

Source note `Navier-Stokes.md` has `\latex-env: equation` and a `$$...$$` block with `\begin{aligned}...\end{aligned}` inside. Expected: a single equation number for the whole block.

$$\begin{aligned}\rho \left(\frac{\partial \vec{u}}{\partial t} + (\vec{u} \cdot \nabla) \vec{u} \right) &= -\nabla p + \mu \nabla^2 \vec{u} + \vec{f} \\ \nabla \cdot \vec{u} &= 0\end{aligned}\tag{1}$$

As described in Equation 1, conservation of momentum couples the pressure gradient to friction. The reference also works with custom text — see the conservation law — and stays clickable as a hyperref.

Source note `Euler-Identity.md` has a simple `$$...$$` block. Expected: “Equation 2”.

$$e^{i\pi} + 1 = 0\tag{2}$$

From Equation 2 the link between the fundamental constants follows directly.

$$\begin{aligned}\rho \left(\frac{\partial \vec{u}}{\partial t} + (\vec{u} \cdot \nabla) \vec{u} \right) &= -\nabla p + \mu \nabla^2 \vec{u} + \vec{f} \\ \nabla \cdot \vec{u} &= 0\end{aligned}\tag{3}$$

If the same equation is embedded again, the reference points to the first embed. See Equation 1.

As with tables and figures, wikilinks to non-embedded equation notes pass through as plain text (no crash), so conceptual references to non-rendered notes still work:

Equation-Does-Not-Exist